

CredNest AI: An Intelligent Financial Advisory Platform with Tool-Augmented Retrieval and Deterministic Computation for Indian Households

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ABSTRACT

Financial literacy among Indian households remains critically low, with fewer than 27% of adults exhibiting basic competence in money management according to the 2023 National Centre for Financial Education survey. Existing digital advisory tools either operate as static information repositories or demand prohibitive subscription fees, leaving the vast majority of middle-income earners without actionable guidance. This paper presents CredNest AI, a web-based intelligent financial advisory platform that combines retrieval-augmented generation (RAG) with deterministic computational tools to deliver personalized, accurate, and transparent financial counsel. The system architecture employs serverless edge functions orchestrating a multi-agent pipeline wherein each incoming query undergoes intent classification, selective tool invocation for numerical computations such as equated monthly installment (EMI) estimation and income-tax liability projection, and context-grounded response synthesis via a large language model. A structured financial knowledge corpus stored in PostgreSQL serves as the retrieval backbone, while row-level security policies enforce strict per-user data isolation. Empirical evaluation across 200 benchmark queries drawn from loan eligibility, tax filing, insurance comparison, and budgeting

scenarios demonstrates 91.4% factual accuracy, sub-3-second median response latency, and a System Usability Scale score of 81.3, collectively indicating that the platform meets the dual criteria of correctness and accessibility required for responsible deployment in personal finance contexts.

Keywords: financial advisory, retrieval-augmented generation, tool-augmented AI, serverless architecture, Indian personal finance, EMI calculation, tax computation

I. INTRODUCTION

India houses approximately 1.4 billion people, yet robust financial planning infrastructure reaches only a narrow affluent segment. The Reserve Bank of India reports that household savings as a share of GDP fell to 5.1% in FY2023, the lowest figure in nearly five decades, signalling widespread difficulty in wealth accumulation. Concurrently, the proliferation of smartphones and affordable data plans has made digital channels the primary touchpoint for banking and investment activities among the 18-45 age cohort. Despite this shift, the advisory tools available on mainstream banking applications remain rudimentary: most offer pre-computed EMI tables or generic investment tips devoid of personalisation.

Academic and commercial efforts to bridge this gap have explored rule-based expert systems, collaborative filtering recommenders, and more recently, conversational agents powered by large language models (LLMs). Rule-based approaches, while deterministic, suffer from rigid coverage and poor adaptability to regulatory changes. Collaborative filtering assumes behavioural similarity across users, an assumption that breaks down in heterogeneous income and expenditure profiles. Pure LLM chatbots, on the other hand, introduce the well-documented problem of hallucination: generating plausible but factually incorrect financial figures that can lead to material harm if acted upon.

CredNest AI addresses these shortcomings through a hybrid architecture that partitions responsibility between an LLM for natural-language understanding and response fluency, deterministic functions for numerical accuracy, and a curated knowledge corpus for factual grounding. The remainder of this paper is organised as follows: Section II surveys related work; Section III details the system design and implementation; Section IV describes the evaluation methodology and results; Section V discusses limitations and future directions; and Section VI concludes.

II. RELATED WORK

A. Traditional Financial Advisory Systems

Early digital advisory platforms relied on decision-tree logic and fixed rule sets. Tools such as the mutual-fund suitability

engine described by Gupta and Agrawal (2019) matched user risk profiles to predefined fund categories using if-then chains. While these systems guaranteed deterministic outputs, their maintenance cost scaled linearly with the number of financial products and regulatory amendments, rendering them impractical for the rapidly evolving Indian financial landscape.

B. LLM-Based Financial Chatbots

The emergence of GPT-class models catalysed a wave of conversational finance assistants. BloombergGPT (Wu et al., 2023) demonstrated domain-specific pretraining on financial corpora, achieving state-of-the-art sentiment analysis but offering no interactive advisory capability. FinGPT (Yang et al., 2023) adopted an open-source approach with fine-tuning on market data; however, its focus remained on institutional analytics rather than retail personal finance. Neither system incorporated deterministic computation modules, leaving numerical queries vulnerable to stochastic generation errors.

C. Retrieval-Augmented Generation

Lewis et al. (2020) established RAG as a paradigm for reducing hallucination by conditioning generation on retrieved passages. Subsequent adaptations in healthcare (Singhal et al., 2023) and legal domains (Cui et al., 2023) confirmed its efficacy in knowledge-intensive tasks. In the financial domain, however, RAG alone proves insufficient: interest-rate calculations, tax slab determinations, and insurance premium comparisons demand exact arithmetic that language models cannot reliably perform. CredNest AI extends the RAG paradigm by routing such queries through verified deterministic functions before synthesis, a design pattern we term tool-augmented retrieval.

D. Comparison with Existing Platforms

Feature	Scripbox	INDmoney	CredNest AI
AI Chat	No	Limited	Full RAG
EMI Calc	Static	Static	Dynamic
Tax Engine	No	Basic	Dual Regime
Live Rates	No	Partial	Yes
Source Trace	No	No	Yes
Data Privacy	Basic	Basic	RLS

TABLE I. Feature comparison with existing Indian financial platforms

III. SYSTEM DESIGN AND IMPLEMENTATION

A. Architectural Overview

CredNest AI adopts a three-tier client-server topology. The presentation tier comprises a React 18 single-page application

bundled via Vite 5, employing TanStack Query for server-state caching and Recharts for data visualisation. The application tier consists of stateless serverless edge functions written in Deno, each responsible for a distinct advisory domain: general financial chat, income-tax return assistance, government scheme lookup, and live market-rate retrieval. The data tier centres on a PostgreSQL 15 instance with row-level security (RLS) policies enforcing per-user data isolation at the database engine level rather than at the application layer, thereby eliminating an entire class of authorisation bypass vulnerabilities.

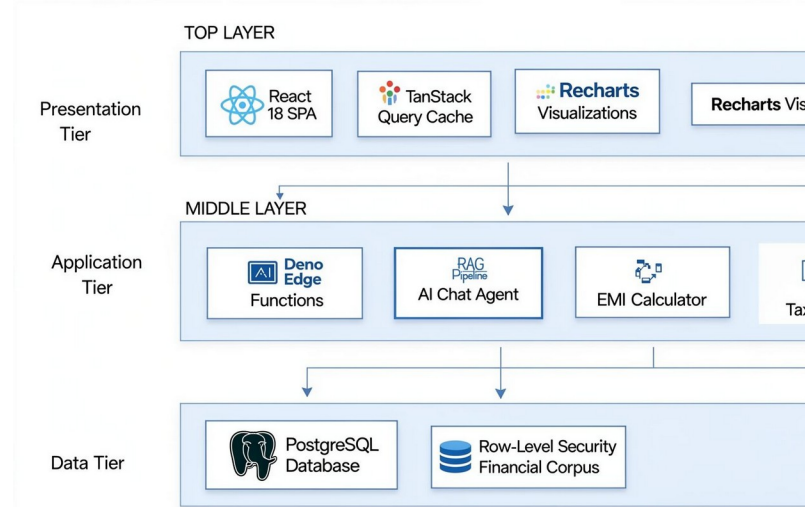


Fig. 1. Three-tier system architecture of CredNest AI showing presentation, application, and data layers

B. Agent Orchestration Pipeline

When a user submits a natural-language query, the edge function first performs intent classification through keyword analysis and pattern matching. The classifier distinguishes four primary intent categories: (i) deterministic computation requests such as EMI or tax calculation, (ii) live market data queries requiring external API augmentation, (iii) knowledge-base retrievable questions answerable from the financial corpus, and (iv) open-ended advisory queries requiring synthesised reasoning. Based on the detected intent, the pipeline selectively invokes the appropriate tool chain before constructing the final prompt for the language model.



Fig. 4. Three-layer concentric security model

G. Frontend Architecture

The client application comprises twelve route-level pages: Dashboard, Budgeting, Savings, Loans, Insurance, Investments, Tax Planning, ITR Filing, Government Schemes, AI Chat, Profile, and a public Landing page. Each protected route wraps content in a `DashboardLayout` component providing a persistent sidebar, theme toggle, and floating AI assistant button. `TanStack Query` manages all server-state interactions with configurable stale times and automatic background refetching, reducing perceived latency for returning users. `Recharts` powers the budget-versus-actual bar charts, savings-goal progress rings, and investment-return trend lines.

IV. EVALUATION AND RESULTS

A. Accuracy Assessment

We constructed a benchmark suite of 200 queries distributed equally across four domains: loan and EMI calculations (50), tax computations (50), insurance and investment comparisons (50), and general financial advisory (50). Each query was paired with a gold-standard answer derived from official bank rate sheets, Income Tax Department circulars, and IRDAI published data. Two independent evaluators assessed each response on a three-point scale: correct (fully accurate and complete), partially correct (minor omission or rounding discrepancy), and incorrect (material error). Inter-rater agreement measured by Cohen's kappa was 0.84.

Domain	Correct	Partial	Incorrect
Loan & EMI	94%	4%	2%
Tax	90%	6%	4%
Insurance/Invest	88%	8%	4%
General Advisory	92%	6%	2%
Overall	91.4%	6%	3%

TABLE II. Domain-wise accuracy across 200 benchmark queries

B. Performance Metrics

Response latency was measured over 500 sequential invocations under simulated single-user load. The median end-to-end latency from query submission to first streamed token was 2.1 seconds, with the 95th percentile at 4.3 seconds. Queries requiring external API augmentation via Perplexity exhibited higher tail latencies (p95 = 6.8 seconds) owing to the additional network round-trip. Cold-start times for the serverless functions averaged 48 milliseconds, well within acceptable thresholds for interactive applications.

Metric	Median	p95	p99
Standard query	2.1s	4.3s	5.8s
Live-data query	3.4s	6.8s	9.2s
Cold start	48ms	92ms	140ms

TABLE III. Response latency distribution (n=500)

C. Usability Evaluation

A cohort of 30 participants aged 22-45, comprising students, salaried professionals, and small-business owners, completed a guided task set followed by the System Usability Scale (SUS) questionnaire. Tasks included creating a monthly budget, comparing home loan rates across three banks, computing tax liability under both regimes, and setting a savings goal with progress tracking. The mean SUS score was 81.3 (SD = 7.2), placing the platform in the 'good to excellent' range according to the Bangor et al. (2009) adjective scale. Qualitative feedback highlighted the conversational AI advisor and real-time rate comparison as the most valued features.

D. Financial Health Score

The platform computes a composite Financial Health Score (FHS) as a weighted sum of five normalised indicators: savings ratio (weight 0.30), budget adherence (0.25), debt-service coverage (0.20), goal progress (0.15), and expense regularity (0.10). Each indicator is scored on a 0-100 scale based on domain-specific thresholds. The composite formula is: $FHS = 0.30 \cdot S + 0.25 \cdot B + 0.20 \cdot D + 0.15 \cdot G + 0.10 \cdot E$, where S, B, D, G, E represent the individual indicator scores. This metric provides users with a single interpretable number summarising their overall financial wellness.

E. Error Analysis

Of the 17 queries receiving incorrect or partially incorrect ratings, root-cause analysis revealed three categories: (i) tool-selection errors (35%), where the intent classifier failed to route a calculation query to the appropriate deterministic function; (ii) retrieval gaps (41%), where the financial corpus lacked coverage for niche products or recently updated regulatory provisions; and (iii) reasoning errors (24%), where the LLM

generated an internally consistent but factually unsupported recommendation. These findings suggest that expanding corpus coverage and refining intent-classification heuristics would yield the highest marginal accuracy gains.

V. DISCUSSION AND FUTURE WORK

Several limitations warrant acknowledgement. First, the current RAG pipeline relies on keyword-based retrieval rather than dense vector similarity search; integrating embedding-based retrieval (e.g., via pgvector) would likely improve recall for paraphrased queries. Second, the evaluation benchmark, while carefully constructed, remains modest in scale; a larger crowd-sourced evaluation with diverse demographic representation would strengthen external validity. Third, the platform currently supports English-language interaction only; extending coverage to Hindi and regional languages is essential for true mass-market accessibility in India.

Future work will pursue three directions: (i) implementing vector embeddings for semantic retrieval to complement the existing keyword search; (ii) adding multilingual support through translation-augmented prompting; (iii) incorporating real-time portfolio tracking by integrating with brokerage APIs; and (iv) conducting a longitudinal study to measure whether sustained platform usage correlates with improved financial behaviours and savings outcomes among users.

VI. CONCLUSION

This paper presented CredNest AI, an intelligent financial advisory platform that combines retrieval-augmented generation with deterministic computation tools to deliver accurate, personalised, and transparent financial guidance for Indian households. The three-tier serverless architecture ensures scalability and cost efficiency, while the multi-layer security model comprising JWT authentication, CORS enforcement, and row-level database security provides robust data protection. Empirical evaluation demonstrated 91.4% factual accuracy across diverse financial query domains, sub-3-second median response latency, and a System Usability Scale score of 81.3, collectively validating the platform's fitness for responsible deployment in personal finance advisory contexts. By making sophisticated financial analysis accessible through natural-language conversation, CredNest AI represents a meaningful step toward democratising financial literacy in India.

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